

Propane Dehydrogenation Process

Report 1

CHE 397- Senior Design II

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Project Charter

Title	Goal
Report #1: Propane Dehydrogenation Process	To optimize the production of propylene using the propane dehydrogenation process with a production of 600 KMTA.
In Scope	Out of Scope
- Design of propane dehydrogenation via commercial OLEFLEX unit - Process Description - Equipment Description - BFD and PFD - Mass Balance	- Demand of propylene - Process feasibility - Location of production - Current prices and trends - Safety concerns
Deliverables	Team Members
- Design plan - Market Analysis - Solution - Competing Processes - Block Flow Diagram - Process Flow Diagram - Material Balance	- Ahmad Abusubaih - Kathya Chavez - Jon Chiu - Yier Li - Marium Murtaza

Task Delegations

Report #1 Design Plan					
Task ID	Task	Start Date	Duration (days)	Finish Date	Assignee
Logistics		27-Jan	3	29-Jan	All
1	Task Delegation	27-Jan	3	29-Jan	All
2	Project Charter	27-Jan	3	29-Jan	All
3	Design Plan	27-Jan	3	29-Jan	All
4	PFD/BFD research	27-Jan	3	29-Jan	Jon
5	Market Analysis Price Research	27-Jan	3	29-Jan	Ahmad
6	Competing Processes Research	27-Jan	3	29-Jan	Yier
7	UOP Oleflex™ Process Re-search	27-Jan	3	29-Jan	Marium
Execution		29-Jan	4	01-Feb	All
8	Introduction	29-Jan	4	01-Feb	Kathya
9	Market Analysis	29-Jan	4	01-Feb	Ahmad
10	Process Description	29-Jan	4	01-Feb	Marium & Yier
11	Appendix 1-4	29-Jan	4	01-Feb	Jon
Wrap-up		01-Feb	1	01-Feb	All
12	Section Unification	01-Feb	1	01-Feb	All
13	Proofreading	01-Feb	1	01-Feb	All

Introduction

Developments in performing the Oleflex™ process has shown promise in the growing Carbon World. In reaches of creating a highly competitive material that has shown useful products to high-quality material. The Oleflex™ processes have many competitors, yet with information researched to creating the best quality material in the market of catalytic dehydrogenation of Propane propylene. With the developing world-changing to cleaner energy, creating a process where the most yield is created for all carbon-based desired material is in high demand.

Studying the processes for Propane Dehydrogenation (PDH) at the primary level changes the chemical structure. Propane is converted to propylene, from a three-chain carbon with eight hydrogens to a carbon skeleton with six hydrogens. By name, propane is made to propylene or methyl ethylene, which are two names given for one propene chemical formula and compound. The break-off of this reaction creates valuable material of byproducts of mostly hydrogen and propylene.

The Oleflex™ process uses PDH and has the highest conversion rate for fuel goal purposes. The importance of this process focuses on the supply and demand for fuel. Finding the best process is the goal and optimizing the ideal percentages while finding the maximum process energy balance at safe regulation for a chosen location in Texas. A simple chemical reaction, yet a complicated system in consideration of the engineering possibilities.

Considering the field service development of the Oleflex™ process, the goal is to optimize the product, cost, and emission reduction and considering the reaction separation for hydrogen and propylene optimizing correct temperatures, pressures, and process controllers in an AS-PEN simulation. The return on investment (ROI) is most crucial for a competing process; the

Oleflex™ process has the best quality of the material. Using every source of free energy, recovery of heat, gas recycling, and material composition helps strategize to creating the most from this process. Making this process more carbon footprint friendly will surpass any competing process.

In this paper, the problem of meeting the gap in the propylene market requirement. The based case unit design proposal with 600 K metric tons per annum (MTA) propylene production. The available economic feedstocks which considered as propane. For the purpose of this work, the united states Taxes is chosen as the desired location of a feedstock source of plant assumed is built. A design plan description analysis is carried out to answer the following question: Which is the desire process flowsheet based upon the problem of the propylene market? What are the most benefited plant and environment-friendly to an overall sustainability performance of the process?

Market Analysis

The proposed process we will be studying is the propane dehydrogenation process. Since our predominant desire is to produce propylene, the propane dehydrogenation process is the leading process in accomplishing this task. Propylene is a plastic that is widely considered one of the most versatile plastics used in almost every subsection of consumer goods. However, there is expected to be a huge “propylene gap,” meaning there is a vast difference from how much demand there is of propylene in the universe vs. what resources can supply of propylene. As shown in Figure 1, in 2021, the propylene gap is expected to be at its highest and continues to grow at a rapid rate. The need for propylene is mandatory, but there is not enough propylene being made. As the population is increasing- there is a high demand for propylene to help aid in providing textiles, consumer products, films, packaging, and technical necessities such as car bumpers. The

reason for this shift is because many steam crackers have been shifting to lighter feedstocks that produce less propylene, and flat demand for gasoline in some regions has limited the amount of propylene produced during oil-refining processes. [1] In order to fill the gap, on purpose technologies such as the propane dehydrogenation process is used.

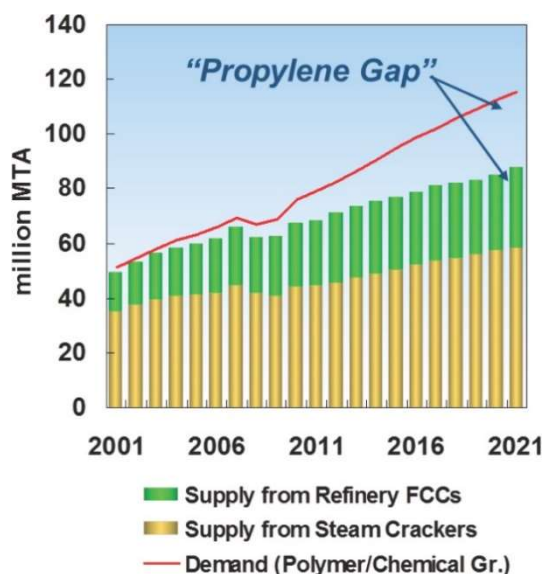


Figure 1 Growing Gap Between Propylene Supply and Demand [1]

Value Chain

Propylene can be used in appliances, electronics, plastic, luggage, toys, wires, cables, piping, and more. As shown in figure 1, this value chain provides an overview of the origin and the usage of propylene [16]. Originally, propylene production was used by other industries for petroleum refining and olefin cracking. However, that has changed over the years due to the demand of propylene in Asia and the abundant US propane supplies. As a result, there are three new propane dehydrogenation plants at the US Gulf Coast, in which are expected to produce 4.3 billion pounds per year. The new PDH plants will help close the gap between the world demand of propylene and the decline of the by-product production [17].

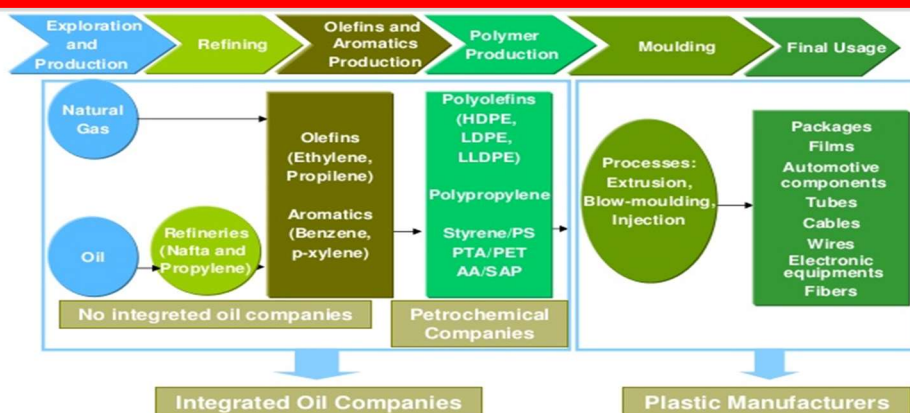


Figure 2 Value Chain of Propylene [16]

Market Drivers

As shown in figure 2, propylene has a demand that is foreseen to grow as the years go by [18]. It is estimated that the demand will grow from 176 million tons to 290 million tons from the years 2011 to 2025. The growth in propylene demands are greater by 3.8% a year. This growth is driven by the great use of propylene in the Asian regions. The consumption of propylene in China is accounted for more than 15% of the demand of the world. The demand of propylene in China is growing to about 5% to 6% per year [17].

U.S. Propylene Market, By Application, 2014-2024 (in BN USD)

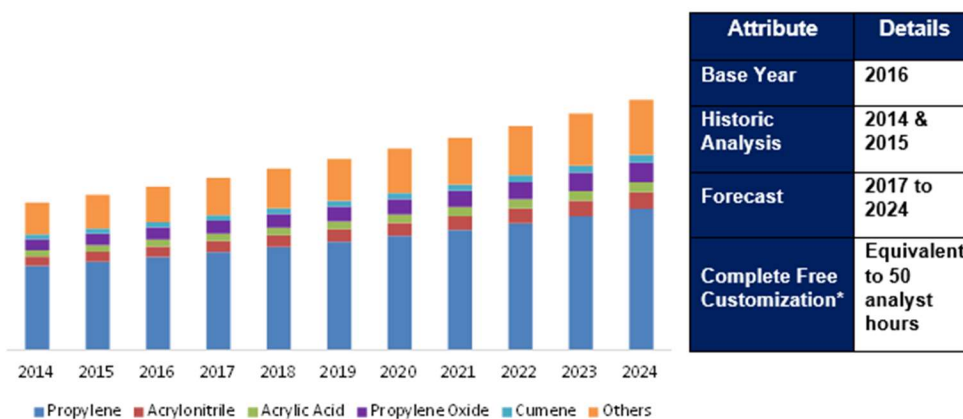


Figure 3 Propylene Market [18]

The highest level of US inventories of propane and propylene has reached 97.7 million barrels [17]. Mainly the increase was in the Gulf Coast regions as highlighted in red for PADD3 in figure 3. Therefore, Texas is a good region to start a production location for propylene. Propane will be readily available to use to produce propylene. By having more propane, the ability to store, transport, and export these commodities will increase as well. The development of tight oil and shale gas remains the main reason for propane production growth [19].

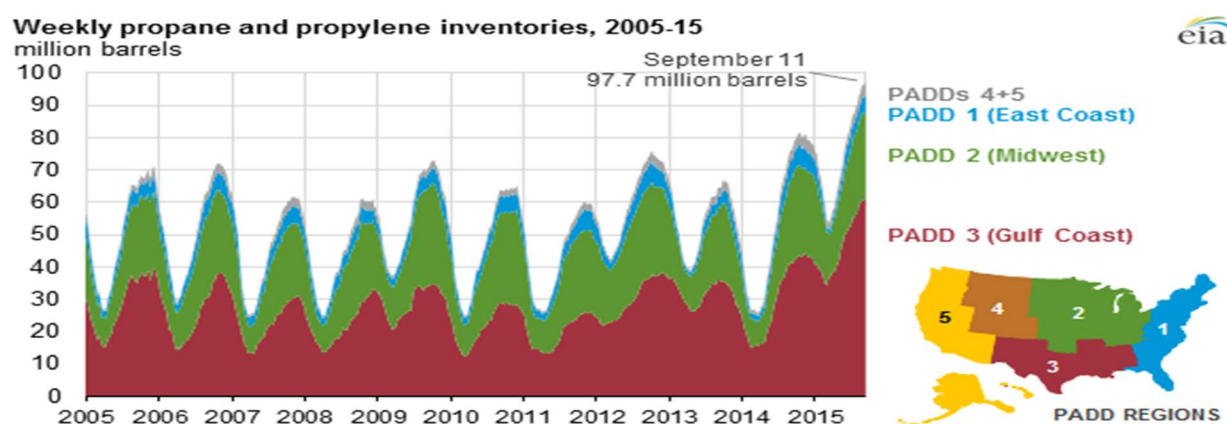


Figure 4: Propane and Propylene inventory [17].

Feedstock

There are many ways to produce “on-purpose” propylene. Among other technologies, propane dehydrogenation is known to be the most economical and shortest route to produce propylene and is commercially available. It is a favored way to produce it in today’s market conditions due to propane being a cheap and in an abundant amount particularly in the US. The prices of propane and propylene are the major aspect that is relied on for profitability. In recent years, there has been a shortage of propylene which increased its prices [17]. The price of propylene is about 40.5 cents per pound [11]. At the same time, prices of US propane are low due to the low demand, constrained exports and abundant shale related output. The price of propane is 5.25 cents per pound [12].

Competitor

One of the top competitors to UOP's Universal Oil Product propane dehydrogenation process is a process called CATOFIN and STAR. These companies produce similar propylene product; however, the company's process was not able to compete with UOP's process. Therefore, the CATOFIN company became bankrupt and the STAR process lacks catalytic regeneration. On the other hand, UOP had high conversion and high selectivity for conversions of paraffins to olefins [13].

Regulatory Environment

At normal temperatures, propylene is a gas and will partition in the air when released. It has a low potential to bioaccumulate, while also being biodegradable [15]. Therefore, propylene is not a hazard to the environment which put no restriction limits on it from the government. Propane is a clean-burning fuel and low-carbon fuel, which can help reduce the harmful air pollutants and greenhouse gases that other conventional fuels may be producing in the environment [14]. However, during the process in the reactor unit. Due to the hydrogenation of propane and regeneration of catalyst required a high temperature which produces a significant amount of coke.

Process Description

Since there is a tremendous demand for propylene in the following years as trends show, the main objective is the desire for this project is to produce as much propylene from a single process. The propane dehydrogenation process (PDH) is a promising catalytic technology that can produce high purity and quality propylene from a propane rich stream- making this a leading process in successfully producing the primary product, propylene. The dehydrogenation of propane to make propylene is the main reaction that occurs in the reactors. Additionally, there are smaller byproduct reactions in the process to remove the hydrogen produced from the reaction

and other side cracking reactions that occur. The primary and side-reactions in the PDH process are shown in Table 1. To achieve a reasonable conversion of propane to propylene the PDH reaction, a Pt-Sn (platinum-tin) catalyst is used inside low pressure and high-temperature reactors.

Table 1. Kinetic Reactions for PDH Process

Primary reaction: dehydrogenation reaction	
$\mathfrak{R}_1$	$C_3H_8 \leftrightarrow C_3H_6 + H_2$
Byproduct reaction: side cracking reaction	
$\mathfrak{R}_2$	$C_3H_8 + H_2 \rightarrow C_2H_6 + CH_4$
$\mathfrak{R}_2$	$C_3H_8 \rightarrow C_2H_4 + CH_4$
$\mathfrak{R}_3$	$C_2H_4 + H_2 \rightarrow C_2H_6$
$\mathfrak{R}_4$	$C_3H_6 + H_2 \rightarrow C_2H_4 + CH_4$

Competing Processes and Technologies

There are many alternative solutions to produce propylene, and the best method can be defined as the process that yields the highest amount of propylene in regular operating conditions. Overall, there are two alternative processes that are great competitors to the propane dehydrogenation process for the production of propylene: the methanol to olefins process (MTO), and the high severity fluid catalytic cracking (HS-FCC). These two processes have the breakthrough technology, which produces high concentrations of propylene and has been used competitively with the propane dehydrogenation process to produce propylene by many facilities. The MTO process is a very renowned and illustrious process which uses an extraordinary catalyst with an active ingredient called sample 34. The process efficiently converts methanol into ethylene and

propylene at very high selectivity and minimal byproducts- allowing the process to produce valuable light olefins very economically and at very high yields. That being said, the strengths of the MTO process is that it helps fill the propylene gap, opens a new path to ethylene and propylene by a low cost of production, generates a high yield of ethylene and propylene, and has a single train capacity of 1.8 million of light olefins. [3] Even though the advanced MTO technology is perfect in supplying chemicals that are needed for consumers and are very high in demand, there are many weaknesses to this process. Firstly, the MTO process makes propylene, but it also makes an abundant source of ethylene which is undesired in our project. The main chemical in consideration is propylene, and the MTO feed is used to make almost equal amounts of ethylene and propylene instead of solely producing propylene. This lowers the overall production of propylene from the same feed stream. In figure 3 below, looking at more prominent technologies for making propylene, the propane dehydrogenation process and the MTO process are seen. By inspecting the trend between the two, it is clearly shown that the MTO process lacks in providing the highest yield for propylene as it produces both ethylene and propylene with limiting propylene to ethylene ratios. On the other hand, the propylene dehydrogenation process selectively produces propylene as the more important product with very few byproducts- ultimately providing the highest propylene yield. Moreover, another weakness of MTO is that the feedstock of the MTO process is made up of mostly coal and natural gas. In the present and future projects, many companies are determining ways to become more environmentally friendly and limiting the use of coal in their new startups. This is also a weakness in terms of our project since more environmentally friendly options are being considered in this project, and the use of coal does not contribute to a small environmental footprint.

Technologies for Making Propylene

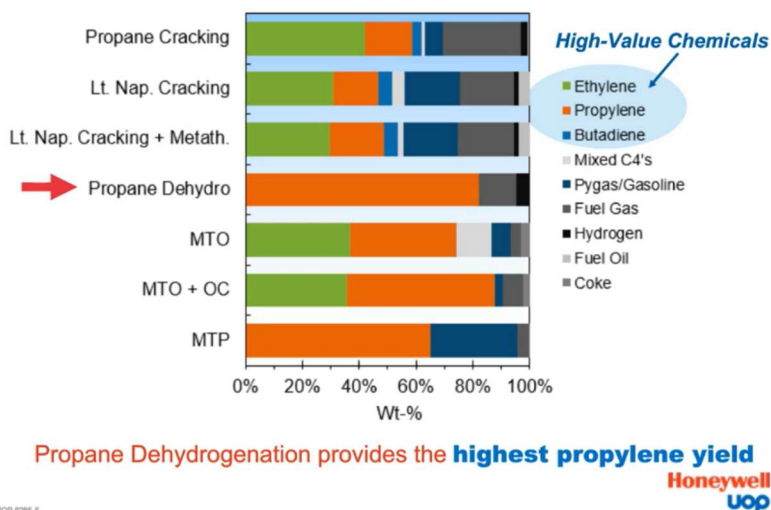


Figure 5 MTO Process vs. Propane Dehydrogenation Process Yield Results [3]

Another competitive process that provides an alternative solution to product propylene is the high-severity fluidized catalytic cracking process (HS-FCC). This process allows refineries to produce petrochemicals from heavy oils by converting a low-value refinery stream into high-value products, suitable for integrated processes. [3] The process's main operating features are the down flow reactor system, high reaction temperature, short residence time, and high catalyst-to-oil ratio with a modified separator system. [3] Some strengths of this process are that its' unique down flow reactor system in the HS-FCC process ensures plug flow without back-mixing. This allows more selectivity toward light olefins. [3] Moreover, HS-FCC produces four times more light olefins (propylene and butene) and higher-octane gasoline than conventional FCC units. [3] There have been many units that are being successfully used at this moment with various feedstocks and feed capacities. One strong weakness of the HS-FCC unit is that even though it aids in producing propylene compared to other conventional FCC processes, it still

does not exceed the yield advantages from the propane dehydrogenation process. Other weaknesses of the HS-FCC unit in comparison to the propane dehydrogenation process is that the HS-FCC process achieves maximum propylene production only when it is operating at high temperatures, short residence times, and introduces horrendous contributions of dry gas and coke formation on the catalysts due to thermal cracking. [3] By looking at figure 4 below, the comparison of the HS-FCC to the propane dehydrogenation process is very noticeable. The propane dehydrogenation still appears to have a favorable propylene yield with minimal by-byproduct. On the other hand, the HS-FCC process has very little contribution to the propylene yield, producing mixed butadiene, gasoline, and few amounts of ethylene as well.

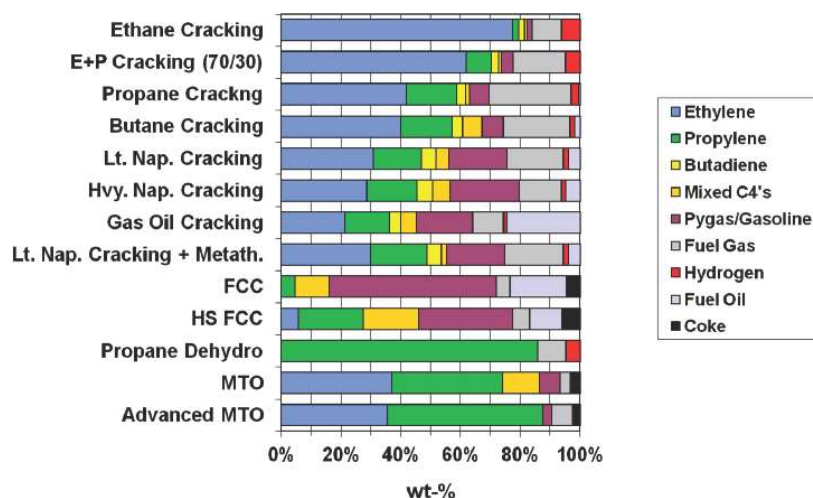


Figure 6 Varying Yields of Propylene by Different Processes [1]

Thus, there seems to be a significant advantage that the PDH process would be the process to form propylene. The different operating conditions, such as high temperature and low pressure and utilized catalysts, and process performance. The reaction schemes used in the mass balance is mainly based on table 1 above, and the side reaction is based on the inner reactor compositions. [7] PHD reaction is highly endothermic since the reactant propane absorbs an abundance of energy from the reactor. Nowadays, (1) the Oleflex™ (moving-bed) from UOP, (2)

CATOFIN (fixed-bed) from Lummus Global and (3) Steam active reforming (STAR) from ThyssenKrupp, (4) Fluidized bed dehydrogenation from Snamprogetti and (5) PDH from Linde-BASF-Statoil are the five-world famous proven commercialized PDH technologies. From these five technologies, the Oleflex™, CATOFIN, and STAR processes are well known and have been proved to be the top performing in propane dehydrogenation in the industry.

The CATOFIN dehydrogenation technology is famous since the process can be operated to a countercurrent reactor and result in the maximum propane conversion with the very high propylene yield. However, the catalyst used in the CATOFIN process is extremely toxic when the catalyst is blowing in hot air and this leads to the loss of catalyst activity from the coke formation and chromium reduction. Moreover, this causes merely an average of two or three years the catalyst activity, resulting in a shutdown of the entire plant to change the catalyst. The STAR process is a better process compared to CATOFIN since it has a higher reliability and availability on stream. The independent parallel reactor trains help convert the propane more efficiently and in a less time-consuming manner. [10] Moreover, its wide operational window makes trips less likely to occur which can be a huge advantage. However, like the Oleflex™ process, the STAR process has major disadvantages that ultimately compelled the decision to examine other processes instead. Some of these weaknesses include a longer start-up and overall shut-down time during the event of a reactor trip. Compared to UOP's Oleflex™, the STAR process takes more time to get the process running again since it does not have on stream availability to allow it to have no short restarts. Unfortunately, there are 'hot stand-by' sequences that make it have a short restart. Also, STAR process has no moving catalyst and uses a fixed bed reactor which leads to less optimization than the Oleflex™ process which has a more unique and uncomplicated reactor process that allows it to continuously run [10] Overall, Oleflex™ dehydrogenation technology is

one of the most used processes which are based on platinum-based (DeH-14) catalyst that is developed in 2002 to promote the dehydrogenation to the maximum size of the plant. Using platinum-based (DeH-14) catalyst prevents the shut down for the catalyst exchanger, and keeping operation continuously running. Oleflex™ technology has various advantages and optimization features that can continuously steam production of propylene. Furthermore, the process itself, provides a low operating cost and low feedstock consumption of the plant and the fastest, continuous process brings the highest money return to compare with the STAR and CATOFIN technologies. [8]

Chosen Process

The solution to obtain the highest production of propylene is the propane dehydrogenation system using UOP's Oleflex™ unit. The UOP process system equipment includes pressure swing Absorption (PSA), moving-bed reactor, CCR unit, distillation trays, heat exchanger, and process instrumentation controls (PIC). This process converts one single feed of mostly propane to one primary product, propylene. It also gives the option to export the hydrogen that accumulates as a byproduct. This process has three main sections- the reactor section, product recovery section, and the catalyst regeneration section. The Oleflex™ process begins with propane entering the Depropanizer where the C4+ and heavier carbon components are separated from the lighter carbons (C3 and less). The propane is then sent to the Oleflex™ reactor unit, which consists of four dehydrogenation reactors and four inter-stage heaters and a reactor feed effluent heat exchanger. [5] The dehydrogenation reactor unit is where propylene and hydrogen are formed. After the platinum catalyst is transferred through all four reactors, it goes through UOP's continuous catalyst regeneration technology (CCR) which burns coke over the catalyst and returns the

regenerated catalyst to the first reactor. [2] To make the product of the reactor unit, the Continuous Catalyst Regeneration (CCR) system is applied, which continuously spends the DeH-14 catalyst and maintains a reasonable conversion between 33% - 40% and approximately 90 % of selectivity. The reactor effluent will be a mixture of propylene, unconverted propane, some light ends, and some hydrogen. [5] The reactor effluent passes through multistage compressors and sent to the Cold Box. The Cold Box unit then cools down the reaction effluent to the liquid phase. In the unit, the effluent stream will be compressed and sent to a treatment system to remove moisture and hydrogen sulfide and then goes into a Cold Box unit which recovers the liquid propylene product and separates the hydrogen abundant net gas. A portion of the hydrogen is recycled back into the feed to prevent the catalyst cracking inside of the reactors. [5] The remaining product is sent to a propylene and propane splitter where the propylene product is recovered, and from the bottom of the splitter. The liquified product and hydrogen from the PSA are sent proportionally to a Selective Hydrogenation Process (SHP) reactor that selectively converts the small number of diolefins to propylene to the Deethanizer to rejected lighter components (mainly C2 and less) to burn off. [5] The bottom steam of the Deethanizer is sent to the Propylene-Propane Splitter, where the final product is separated. [9] A simple block flow diagram illustratively describes the process in figure 8 below

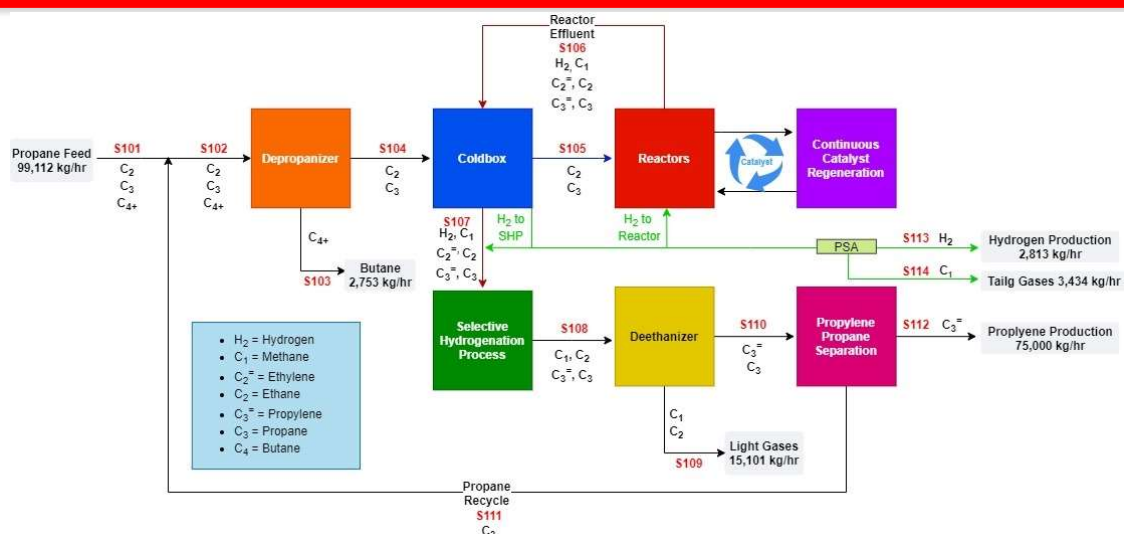


Figure 7 Block Flow Diagram of Oleflex™ Process with Simple Mass Balance

The other reason the Oleflex™ solution was chosen to produce propylene out of all the other propane dehydrogenation solutions is indisputable. The Oleflex™ process has many benefits other than just giving the highest yield of propylene. Firstly, the process's capital expense and operating expense is the lowest of all time. In capital expenses, there is a savings worth of 55 million dollars to 75 million dollars, which is 15% to 20% lower than what the original capital expenses were. [6] Also, the process saves 6 million dollars every year (for a 600 kMTA or 661 kTA unit) in operating expenses. [6] Additionally, the Oleflex™ process has a smaller environmental footprint by significantly lowering emissions of NO_x, NH₃, CO₂, and wastewater. [6] This was a key factor as to why this solution was picked over other processes because eliminating additional sources of environmental damage is very valuable. Moreover, it has been known to have a fast startup (< 36 months) and has been proven to be a safe and reliable technology by many companies. The most crucial reason the Oleflex™ process was chosen is that this process provides up to 10-15 million dollars a year of additional margin from higher onstream availability. [6] This is solely obtained from UOP's Oleflex™ process since it provides the highest plant

productivity. Change the catalyst online without having to shut down the unit. This allows operating units to change out the catalyst without any shutdown requirement, which increases plant productivity, and ultimately the time on stream. [6] In addition, the process uses adiabatic hot wall reactors. There are no reactor liners on the inside of the reactors which are usually susceptible to failures and ultimately shutdowns-this provides more stable operation. [6] Furthermore, from the positive reactor pressure of the Oleflex process, there are no thermal isolation valves that are usually susceptible to leakage and failure. [6] Lastly, the key feature that no other propane dehydrogenation process has other than the Oleflex™ unit is the independent reactor and catalyst regeneration section that allows for continuous operation at steady state conditions without having to switch between the reaction to regeneration modes- creating less thermal stress on the equipment and allowing more time onstream. [6] There is no thinking needed when it comes to comparing the features that the Oleflex™ unit provides from its lowest overall cost of production, the smaller investment, high on-stream availability, and smallest environmental footprint. Today, UOP has been awarded 40 of the last 47 competitively bid propane dehydrogenation projects worldwide since 2011 and currently has its' Oleflex™ process being used in 18 of the 28 operating propane dehydrogenation units in the world. [6]

Appendix 1 Design Basis

1. Measurement System: Metric

2. Equipment Numbering System

AC	Air cooler	G	Grinder, mill	PRV	Pressure relief valve
B	Boiler	H	Heater (fired or electric)	R	Reactor
C	Compressor, blower, fan	J	Ejector, jet, turboexpander	SP	Sample point
CT	Cooling tower	M	Motor	T	Storage tank
D	Dryer	ME	Miscellaneous equipment	V	Vessel (including columns)
E	Exchanger	MX	Mixer		
F	Filter, classifier	P	Pump		

- Equipment will be identified by alphabetic prefix as defined here, followed by three-digit serial number unless otherwise indicated
- First digit - process section
- Second & third digits - equipment count

3. Primary Products [20]

Product Name	Propylene
Product Grade	ACS
MSDS Form Number	CAS # 115-07-1
Production Rate	
Metric Tons per year	600,000
Metric Tons per hour	75
Product Purity (wt%)	99.5
Product Shipment mode	Gas

4. Primary Raw Materials [20]

Feedstock Name	Propane
Feedstock Grade	ACS
MSDS Form Number	CAS # 74-98-6
Feedstock Availability	
Metric Tons per year	792,902
Metric Tons per hour	99.11
Feedstock Price	\$0.1157/kg

Known Feedstock Impurities	
Name	PPM
Methane	993
Ethane	2700
Butane	800

5. Site Information [21]

Location	Freeport, Texas
Low ambient temperature (C)	17.8
High ambient temperature (C)	25.6
High ambient relative humidity (%)	39%
Site elevation (m)	2
Maximum wind loading (kph)	16.0934

Appendix 2 Block Flow Diagram

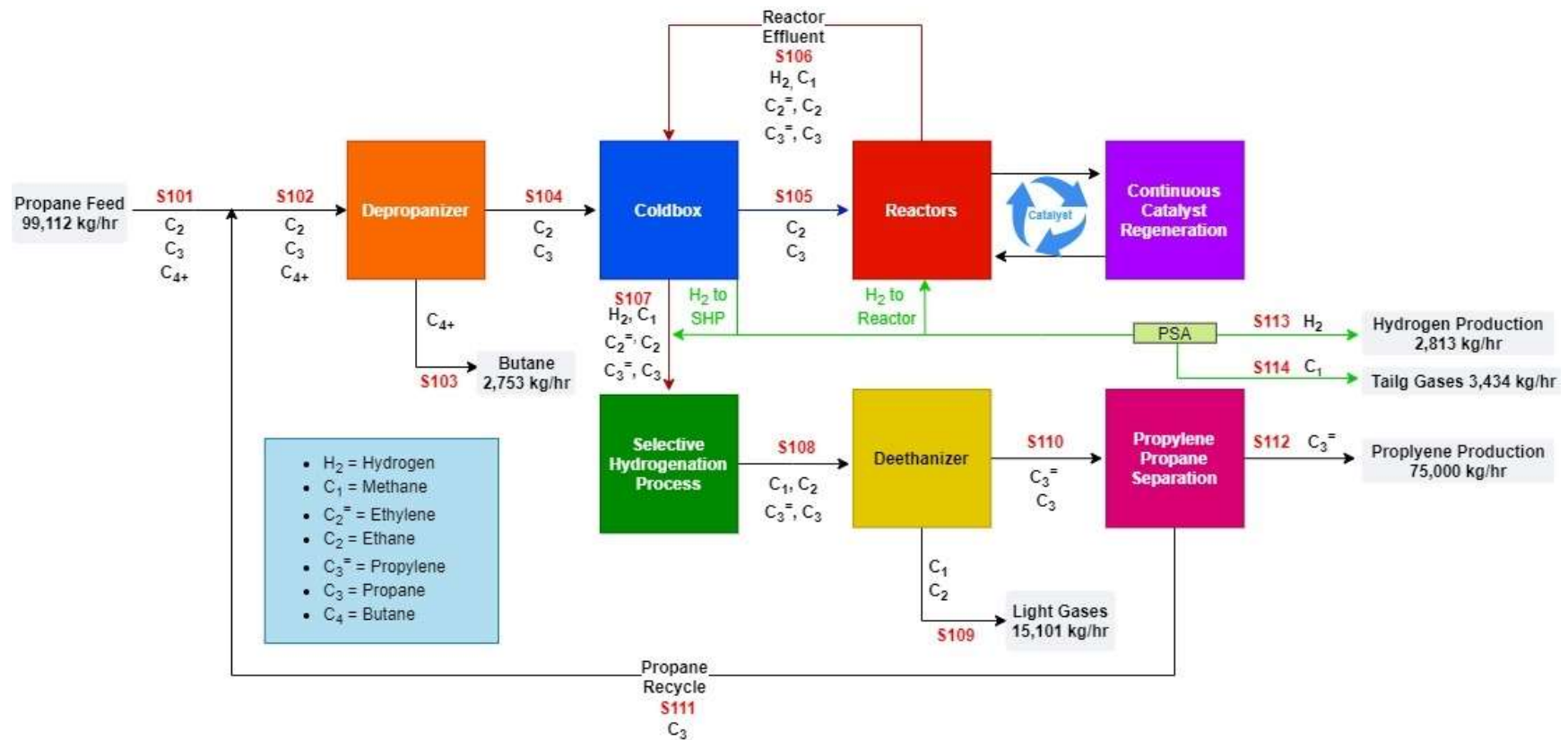


Figure 8 Propane Dehydrogenation Block Flow Diagram with Mass balance

Appendix 3 Process Flow Diagram

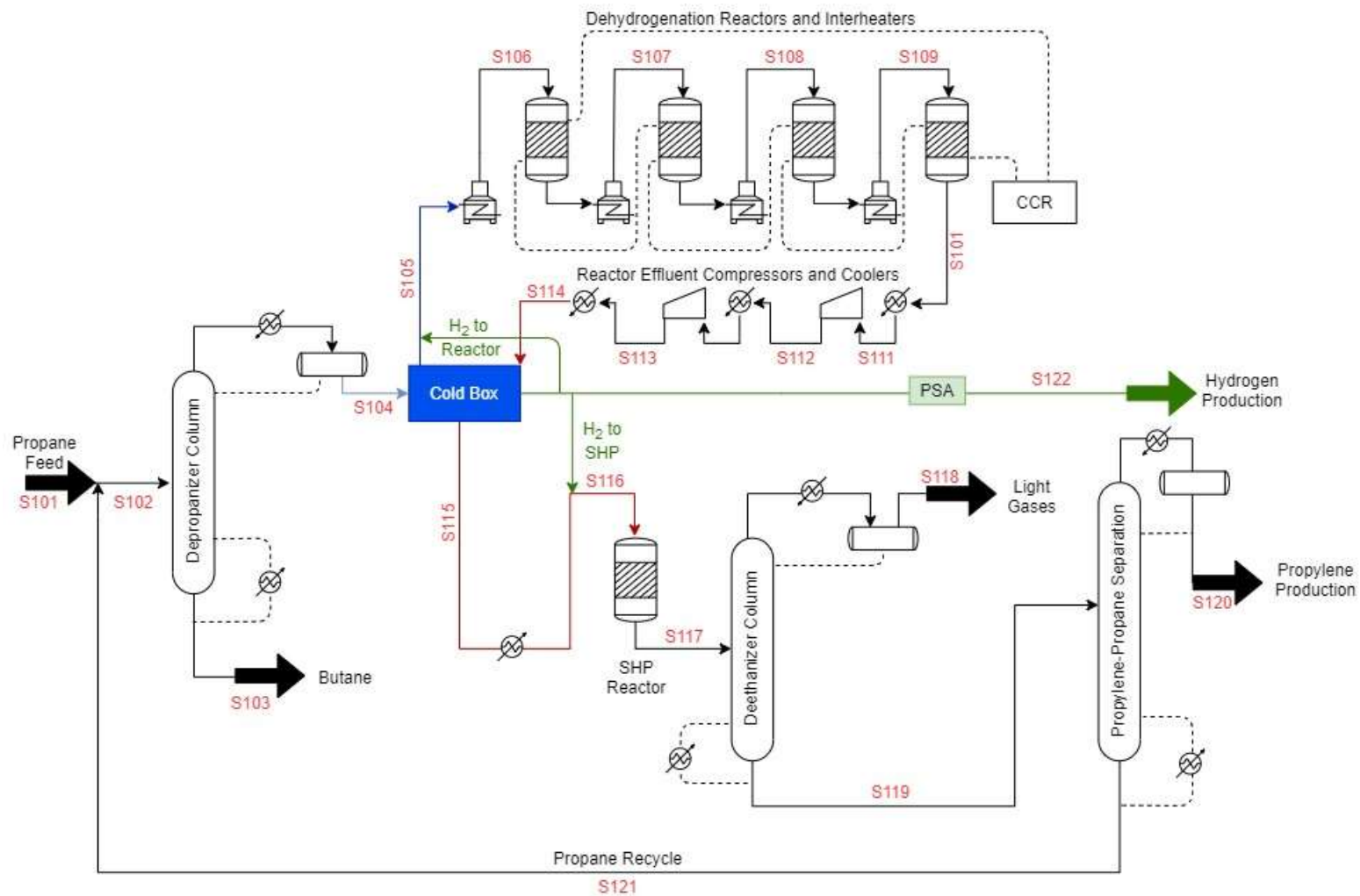


Figure 9 Propene Dehydrogenation Process Flow Diagram

Appendix 4 Mass Balance

Table 2 Propane Dehydrogenation Mass Balance

Mass Flows	Units	S101	S102	S103	S104	H2 to RXN	S105	S106	H2 to SHP
Total	kg/hr	99112.79	269599.36	2753.32	266846.04	36050.30	302896.34	302885.59	669.27
Hydrogen	kg/hr	0.00	0.00	0.00	0.00	18152.11	18152.11	21635.23	669.27
Methane	kg/hr	98.39	98.39	0.00	98.39	16840.38	16938.78	22757.12	0.00
Ethylene	kg/hr	0.00	0.00	0.00	0.00	820.18	820.18	10290.16	0.00
Ethane	kg/hr	1988.05	1988.05	0.00	1988.05	237.63	2225.67	2981.30	0.00
Propylene	kg/hr	0.00	24.63	0.01	24.62	0.00	24.62	73789.69	0.00
Propane	kg/hr	96947.12	267408.26	2674.08	264734.18	0.00	264734.18	171431.30	0.00
Butane	kg/hr	79.23	80.03	79.23	0.80	0.00	0.80	0.80	0.00

Mass Flows	Units	S107	S108	S109	S110	S111	S112	S113	S114
Total	kg/hr	260587.07	260588.44	15101.83	245486.61	170486.57	75000.00	2813.84	3434.38
Hydrogen	kg/hr	669.27	0.00	0.00	0.00	0.00	0.00	2813.84	0.00
Methane	kg/hr	2685.34	2685.34	2685.34	0.00	0.00	0.00	0.00	3231.41
Ethylene	kg/hr	9312.59	0.00	0.00	0.00	0.00	0.00	0.00	157.38
Ethane	kg/hr	2698.08	12681.31	12416.49	264.82	0.00	264.82	0.00	45.60
Propylene	kg/hr	73789.69	73789.69	0.00	73789.69	24.63	73765.06	0.00	0.00
Propane	kg/hr	171431.30	171431.30	0.00	171431.30	170461.14	970.12	0.00	0.00
Butane	kg/hr	0.80	0.80	0.00	0.80	0.80	0.00	0.00	0.00

Assumption and Basis:

- Targeted Production 600 KMTA, 75,000 kg/hr
- Operating Year: 8000 hours
- Conversion C_3 to C_3^- : 33.3%
- Selectivity C_2^- to C_2 : 90%
- Column Separation Rate: 99.5%
- Propane-Propylene Separation: 96.5%
- Hydrogen to Reactor H_2/C_3 Ratio: 1.5
- Hydrogen to SHP Reactor H_2/C_2^- : 1

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